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MILITARY SPECIFICATION

TRACTOR, WHEELED, AIRCRAFT TOWING A/S32U-30, FOR 500,000 to 750,000 POUNDS GROSS WEIGHT AIRCRAFT

1. SCOPE

1.1 This specification covers one type of aircraft towing, wheeled tractor, designated A/S32U-30.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

VV-F-800	Fuel Oil, Diesel

Military

MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-C-3143	Chain Assembly, Tire, for Military and Commercial Tire Sizes
MIL-F-3541	Fittings, Lubrication
MIL-E-5220	Extinguisher, Fire, Aircraft, Type A-20
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-B-11188	Battery, Storage, Lead-Acid

STANDARDS

Military

MIL-STD-12	Abbreviations for use on Drawings and in Technical-Type Publications
MIL-STD-17	Mechanical Symbols
MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

FSC 1740

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MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-281	Automobiles, Trucks, Truck Tractors, Trailers, and Trailer Dollies, Preservation and Packaging of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MS33586	Metals, Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Interstate Commerce Commission

Motor Carrier Safety Regulations

(Application for copies of ICC regulations should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

Society of Automotive Engineers, Incorporated

Standard J695 Turning Ability and Off Tracking

(Application for copies of the SAE Handbook should be addressed to the Society of Automotive Engineers, Incorporated, Two Pennsylvania Plaza, New York, New York 10001.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The tractor shall consist of at least the following components:

Item of System Name	See Requirement
Power train assembly	3.7
Steering system	3.8
Lighting and electrical system	3.9
Crew accommodations	3.10
Miscellaneous provisions	3.11.

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3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3.1 Commercial parts. Commercial parts conforming to SAE standards may be used.

3.4 Materials

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically-sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Protective treatment. When materials used in the construction of the tractor are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall not be used.

3.4.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.4 Wood materials. Wood materials shall not be used in the construction of the tractor.

3.5 Design and contruction. The tractor shall be a commercial item of such design and construction as to meet the requirements specified herein. The tractor shall be designed and constructed for towing a 750,000-pound rated trailed load over wet, dry, or ice and snow covered concrete and macadam surfaces under conditions specified herein.

3.5.1 The tractor shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1.1 The tractor shall be so designed as to be free of interference and obstruction to servicing. There shall be no tendency of the unit to pick up mud, snow, or ice which will interfere with servicing and operation.

3.5.2 Reliability. The tractor shall have a minimum MTBF (mean-time-between-failures) of 543 hours with a minimum reliability of 0.9981 for a 1-hour mission at a confidence factor of 0.90 (see 6.3).

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3.5.3 Maintainability. The tractor shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility

d. Permit maintenance with general-purpose tools and equipment normally available commercially. The use of special-purpose tools and equipment shall be approved by the procuring activity

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolt head sizes to the practicable minimum.

3.5.4 Ground clearance. Ground clearance under the tractor shall be not less than 14 inches or that required to approach, ascend, breakover, and depart a 20° ramp, whichever is greater.

3.5.5 Transportability. The tractor shall be designed and constructed for shipment in an assembled condition as follows:

3.5.5.1 Air. The tractor shall be transportable by air in a Type C-141 aircraft, or equivalent. The tractor shall meet the acceleration forces specified in MIL-A-8421.

3.5.5.2 Rail. The tractor shall be transportable by rail flatcar and shall withstand the forces incurred in bumping the car to result in impact speeds of 11 mph.

3.5.5.3 Lifting rings. The tractor shall be provided with lifting rings located at accessible points that will permit cable attachment and approximate level hoisting without damage to the tractor or its components. The rings shall be so located on the tractor that transportation personnel can rig safe lifting slings from common cable and spreader bar components for lifting by a single-hook, overhead hoisting facility. All lifting rings shall be designed with a minimum safety factor of 4 and an inside diameter of 3 inches. The lifting rings may also serve as tiedown rings.

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3.5.6 Mobility provisions. Suspension system (springs and shock absorbers, if required) vehicle brakes shall be as specified in MIL-M-8090 for type III mobility. Tires shall be 18:00-25, 28 ply. Commercial wheels will be acceptable provided they meet the performance requirements specified herein.

3.5.7 Lubrication. The tractor shall be designed and constructed to operate satisfactorily when lubricated with the standard military lubricants specified in MIL-M-8090.

3.5.7.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. Extended fittings shall be provided to lubricate parts or assemblies that are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.5.7.2 Lubrication chart. A lubrication chart shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricant required for critical temperatures. The chart shall be permanently attached to the tractor in an accessible and convenient location.

3.5.8 Winterization. The tractor shall be so designed and constructed that external or internal winterization aids will not be required for operation down to -20°F. The tractor shall incorporate features that facilitate use of standard ground heaters for preheating purposes. Commercial-type hot water heaters shall be provided for cab heating and defrosting purposes.

3.5.9 Safety features

3.5.9.1 Guards. All rotating or reciprocating parts and all parts subject to high operational temperatures or are electrically energized, that are of such a nature or so located as to be or become a hazard to the safety of operating personnel, shall be insulated, enclosed, or guarded.

3.5.9.2 Nonskid material. All running-boards, steps, and walkways external from the cab shall be of a skidproof design which will permit snow to fall through rather than accumulate.

* 3.5.9.3 Starting safety devices. The tractor shall be provided with a safety device to prohibit engine starting unless the transmission is in a neutral position. The tractor shall also be provided with a frequency sensing electrical device which will prevent engagement of the starter drive when the engine is running. These devices shall not prevent engine starts due to vacuum buildup when cranking the engine under any specified temperature extreme.

3.5.9.4 Captive fill caps. Safety chains shall be attached to all liquid fill caps to prevent their loss or misplacement.

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3.6 Performance. The tractor shall operate satisfactorily under the following conditions:

a. Temperatures ranging from -20° to +125°F

b. Fungus growth as encountered in tropical climates

c. Rainfall as encountered in any locale.

* 3.6.1 Drawbar pull. The fully equipped tractor, having a normal gross vehicle weight of not less than 63,000 pounds and without traction aids, shall develop a drawbar pull of not less than 50,000 pounds. Drawbar pull shall be developed in low gear on a clean, level, brushed, concrete surface at a frictional coefficient of 0.8. This drawbar pull shall be the available force in pounds pull at the tractor pintles and over and above the force required to propel the tractor.

3.6.2 Moving capability. The tractor shall be capable of moving an aircraft weighing 750,000 pounds forward or rearward under the following conditions:

a. Level terrain at a speed of not less than 5 mph

b. Grades up to 3 percent at a speed of not less than 3 mph on clean concrete or macadam surfaces with a frictional coefficient of 0.8.

3.6.2.1 Vehicle speed. The tractor shall be capable of attaining a maximum speed of not less than 25 mph when not towing an aircraft.

3.6.2.2 The fully equipped tractor shall demonstrate a positive capability of providing the following:

a. Not less than 9,000 miles of operation at no load over improved roads at a speed of not less than 20 mph

b. Operation over 4,000 miles under loads as follows:

(1) 200 miles at 50,000 pounds drawbar pull

(2) 1,000 miles at 40,000 pounds drawbar pull

(3) 1,600 miles at 30,000 pounds drawbar pull

(4) 800 miles at 20,000 pounds drawbar pull

(5) 400 miles at 12,000 pounds drawbar pull.

3.6.3 Turn radius. The turn radius of the tractor, measured from the outer extremity of the outside tire, shall be 24 feet or less when turning to the left or right

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in four-wheel steer. Steering brakes shall not be used to meet this requirement. Maximum cramping angle shall be attainable when turning left or right in oblique and two-wheel steer as well as four-wheel steer.

3.6.4 Braking ability. The tractor shall be capable of stopping an aircraft weighing 750,000 pounds within a distance of 35 feet from a speed of 5 mph when the traction coefficient is 0.6. The tractor shall be capable of holding the aircraft when heading up or down a 2.6 percent incline without use of the aircraft braking system.

* 3.7 Power train assembly. The power train assembly, including but not limited to the engine, torque converter, transmission, transfer case, drive lines, axles, wheels, and tires, shall be selected from those available as standard commercial items, previously manufactured, and currently used in commercial or military applications. The capacity of such components shall be based on the stall torque capacity of the torque converter or the slip torque capacity of the tires through the intervening gear reductions with the tractor ballasted to a gross vehicle weight of not less than 63,000 pounds while operating in lowest gear on dry, clean, rough concrete or macadam surfaces with a frictional coefficient of 0.8 (all wheels driving). Power train components of electric and hydrostatic drives are considered acceptable for meeting this specification provided they meet performance and operating characteristics of comparable components described herein. If a hydrostatic drive is furnished to meet the requirements of this specification, forward and reverse movement of the tractor shall occur by reversing fluid flow to the hydraulic motors and gear drives. Chain drives are not acceptable for such purposes. The hydraulic pump shall be the positive displacement type.

* 3.7.1 Engine. The tractor shall be powered with a commercial standard design diesel engine rated for operation with fuel conforming to VV-F-800. Horsepower and torque characteristics shall be sufficient to provide the performance specified herein. Engine manufacturer's published ratings for the engine shall not be arbitrarily raised to meet the requirements specified herein. Engine horsepower shall be provided at an rpm below the manufacturer's published rated maximum which will assure performance with an adequate margin of safety. Engine torque shall be provided at an rpm which will assure performance with an adequate margin of safety. Engine capacity shall be such that the automotive-duty horsepower input required by all items driven by the engine will not exceed 85 percent of the full throttle setting at sea level and at 60°F. An engine overheat relay shall be provided and installed to indicate engine overheat. The relay shall actuate a red light visible to the vehicle driver. The ignition shall be actuated by a keyless switch with not less than 2 positions, legibly marked ON and OFF.

3.7.2 Muffler. A spark arresting muffler shall be provided. The exhaust pipe shall discharge to one side of the tractor or be directed upward by use of a vertical stack. The exhaust shall not be discharged to the immediate front or rear of the tractor.

3.7.3 Transmission. The tractor shall be equipped with a transmission in which a limited number or a single, continuously-variable torque ratio change occurs

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automatically with only instantaneous interruption of power flow. Any further required ratio changes shall be controlled by the operator through full power shifting with only instantaneous interruption of power flow through the drive train. Clutch actuation shall be accomplished automatically by demand-type controls at the transmission output shaft or through positioning of the shift control lever.

3.7.4 Axles. Front and rear axles shall be of the steerable type previously used in commercial applications and entirely suitable for the service specified. The axles shall be so designed that the wheel will not be released in the event of axle shaft failure. Maximum practicable interchangeability shall be maintained between the front and rear axle components. Final reduction shall be in the outer wheel hub and shall be the planetary type.

* 3.7.5 Proportional wheel drive. An automatic means shall be provided for permitting differences in speed between wheels while insuring drive through any wheel proportional to the weight on the wheel, the type surface, and frictional coefficient. The drive shall not disengage while turning, during maneuvering of the tractor, or when backing aircraft.

3.7.6 Transfer case. A two-speed transfer case may be used to provide the airplane towing and road performance specified herein. If a two-speed transfer case is used, synchromesh gears shall be used. The transfer case clutch shall be actuated automatically with positioning of the gear shift lever. Not more than a 1-second delay shall be required for interruption of power flow through the transfer case due to such gear changes.

3.7.7 Tire chain clearance. Adequate clearance shall be maintained around all tires to facilitate use of tire chains. This clearance shall be not less than 3 inches in all directions around the tire. Tire chains shall conform to MIL-C-3143.

3.7.8 Fuel tank. The fuel tank shall have a capacity sufficient to provide a cruising range of not less than 150 miles of operation without refueling.

3.8 Steering system. The tractor shall be provided with a mechanical power assist steering system incorporating features which will permit the operator to turn the wheels while the tractor is stationary with the engine idling with not more than 15 pounds force required at the steering wheel ring. Not more than three turns of the steering wheel shall be required to turn the wheels from maximum cramping angle right to maximum cramping angle left or vice versa. The steering wheel shall be not less than 18 inches in diameter. At least 4 inches clearance shall be provided between the outer edge of the steering wheel and the nearest projection.

3.8.1 Steering and control. The tractor shall have four steerable wheels as required to effect a 24-foot turn radius in either direction. Front and rear wheels shall be steerable in a manner to provide coordinated four-wheel steer, oblique steer, and front-wheel steer only. Maximum cramping angle of each wheel (except rear wheels when in front wheel steer) required to obtain the 24-foot turn radius shall be obtained while in any steering configuration. Turn radius shall be measured to the outside wheel as defined in SAE Recommended Practice No. J695.

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3.9 Lighting and electrical system. The nominal voltage of the tractor shall be 24V. Standard military 24V components shall be used. The battery shall be 12V, selected from MIL-B-11188, and shall be not less than 100-amp hour capacity. The battery shall be mounted under the engine hood or in an easily accessible, readily removable outside compartment which shall be furnished with a quick release cover. The alternator shall be a heavy-duty type that will supply charging current to the battery at engine idle speed and effectively prevent battery discharge (run-down) during all operations, when all lights are being used at extreme low temperatures. All lights, other than the spotlights, shall be recessed or located in a protected position to prevent damage while pushing or towing other equipments or placing wheel chocks or tool boxes aft of the cab. All circuits shall be protected with overload circuit breakers in lieu of fuses. Ground polarity of the system shall be negative. Lighting requirements shall conform to ICC regulations and shall consist of the following:

a. Four sealed beam headlamps, two located in the front and two in the rear in the normal headlamp positions, with high and low beam. The front and rear headlamps shall be controlled by a separate switch. A single dimmer switch shall control the high and low beam of both sets

b. Two combination tail- and stoplamps mounted one on the left rear and one on the right rear of the tractor

c. A floodlamp (in fixed position) on right rear of the tractor to light the towbar

d. Two sealed beam, adjustable spotlamps, one located on the upper rear section of the driver's cab in the center, and the other located on the left front corner post of the cab.

3.10 Crew accommodations

* 3.10.1 Cab. When required (see 6.2), the tractor shall be equipped with an insulated, two-man, enclosed, hard-top cab. The cab shall be heated and ventilated for operation in temperatures from -20° to +125°F. All instruments, controls, seats, and other cab accessories shall be suitable for all weather use when no cab is installed. The cab shall provide for ease of entry and egress to men up to 6 feet 6 inches in height, wearing heavy winter clothing. The cab shall be so constructed that maximum visibility is provided in all directions including front, rear, and sides as required to facilitate coupling, uncoupling, and effective aircraft movement. Adequate windshield heating, defrosting, and wiping shall be provided to facilitate operator visibility during inclement weather. The cab shall be furnished with glare-proof windshield and shall be readily removable from the vehicle. Provisions shall be made for defrosting the rear window of the cab.

* 3.10.2 Towing crew transportation. The tractor shall be designed to provide adequate space aft of the cab for transporting a four-man towing crew and tools. Seating space for each man shall be not less than 19 inches long by 10 inches deep. Space shall be provided to accommodate four tool boxes, each 24 inches long, 12 inches wide,

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by 12 inches high. A tubular guardrail shall be provided as a backrest for the crew and to preclude crew members from being thrown off the moving vehicle. A step(s) for crew entry and egress to and from the rear body shall be furnished if required. A step(s) of not more than 24 inches from the ground to the first step shall determine necessity for steps.

3.11 Miscellaneous provisions

3.11.1 Compressed air devices. Complete air supply devices, such as an air compressor, storage reservoir, low air pressure warning buzzer, fittings, connectors, and all necessary parts and equipment required for an adequate system, shall be provided. The complete system shall conform to ICC Motor Carrier Safety Regulations.

3.11.1.1 Air compressor. The air compressor shall be of the engine-driven and lubricated type having a capacity of not less than 12 cfm at 1,200 rpm.

3.11.2 Extinguishers. Two fire extinguishers conforming to MIL-E-5220 shall be provided and mounted in a location accessible to personnel stationed outside the vehicle. The extinguishers shall be located in a recessed or protected position to prevent damage while pushing or towing other equipment.

* 3.11.3 Tool box and chock compartment. A combination tool box and chock compartment shall be located amidships on the right side of the vehicle approximately 14-3/4 inches above ground. The box shall be approximately 23 inches high, 68 inches long, and may vary in depth from 10 to 16 inches. A hinged door shall be provided for access to the compartment. The door shall cover the full compartment width and height and shall be hinged at the top of the compartment to provide opening through a 180° arc. A suitable hook shall be provided to latch the door to the guardrail when opened. A heavy-duty door latch shall be provided to secure the door in the closed compartment. The latch shall be of sufficient size to allow four-finger operation while wearing arctic mittens.

3.11.4 Pintles. The tractor shall be equipped with two pin-and-clevis type pintles, one front and one rear, approximately 30 inches above the ground when the tractor is ballasted to a gross weight of 63,000 pounds. The pin shall be not less than 2-1/2 inches in diameter. The clevis shall be designed to accommodate a lunette having a 3-inch diameter and constructed of 1-3/4-inch bar stock. A heavy chain shall be secured to the top of the pin and along the clevis body to prevent loss of the pin. The chain shall be adequate to withstand ten 6-foot free falls of the pin being stopped by the chain only without failure.

3.11.5 Winch. Provisions for mounting a winch shall be made on the tractor to facilitate field installation as required. When specified (see 6.2), a power-take-off-driven winch assembly, complete with drum, cable, controls, gear box, and other accessories required for efficient operation, shall be furnished and installed on the tractor. Capacity of the winch shall be not less than 80,000 pounds line pull on bare drum. Winch line speed shall be not less than 30 fpm on bare drum. A safety shear

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pin shall be installed in the winch drive to preclude overload. Capacity of the shear pin shall be +10 -0 percent. Failure of the shear pin shall cause no damage to or misalignment of winch components. When the winch is not specified, counterweights shall be provided to produce the same gross vehicle weight and weight distribution as when the winch is installed.

3.11.6 Gages and controls. Gages and controls shall be properly illuminated for night operation without adverse effect on the operator's vision. The tractor shall be equipped with gages such as an ammeter, speedometer, odometer, hourmeter, tachometer, temperature gages, oil pressure gages, torque converter pressure gage, fuel gage, air pressure gage, and high-low beam indicator.

* 3.12 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.13 Radio interference. The tractor shall be constructed and shielded in a manner that radio interference generated during operation under any conditions specified herein will not exceed the limits established in MIL-STD-826, method 7001.

3.14 Dimensions. Basic tractor dimensions shall be as follows:

a. Angle of approach — Not less than 25°

b. Angle of departure — Not less than 25°

c. Overall width — Not more than 98 inches

* d. Overall height — Not more than 114 inches.

* 3.15 Weight. The individual components of the tractor shall be as lightweight as possible, consistent with good design practices. The assembled tractor shall be ballasted to result in a gross vehicle weight of not less than 63,000 pounds.

3.16 Finishes and protective coatings. Cleaning, painting, and finishing shall be in accordance with MIL-STD-808. The exterior color shall be yellow, film designation DG.

3.17 Markings

3.17.1 Transportation data plates. Transportation data plates sized and marked as required in MIL-P-514 shall be provided. All other data plates, placards, charts, instruction plates, et cetera, shall conform to MIL-P-514, of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12. Symbols shall be in accordance with MIL-STD-17.

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3.17.2 Operational markings. The operational markings shall be applied in accordance with the marking requirements of MIL-STD-808. Stenciling shall be in accordance with MIL-STD-808.

3.17.2.1 Tire pressure. The tire inflation pressure shall be stenciled on the tractor as near each wheel as practicable.

3.17.3 Identification of controls. All valves, gages, controls, and indicators, other than standard vehicular instrumentation, used in the operation of the tractor shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light, et cetera, will not cause them to fade or become eradicated. Tags or decals shall not be used.

3.17.4 Operating instructions. Where applicable, brief operating and precautionary instructions shall be permanently affixed on or near the control panel. The instructions shall be clear, concise, and adequate to enable operation without damage to the unit or injury to operating personnel. Instruction panels shall be of sheet aluminum or sheet zinc not less than 0.050 inch thick, engraved or etched to produce raised or depressed markings with a black or other dark color background, and with a border of not less than 1/4 inch.

3.17.4.1 The following instruction plate shall be installed in a conspicuous location on the dash panel of the cab:

CAUTION

THIS TRACTOR IS EQUIPPED WITH ELECTROMAGNETIC INTERFERENCE SUPPRESSION DEVICES. CARE SHOULD BE EXERCISED DURING MAINTENANCE TO RETAIN AND REINSTALL THESE DEVICES PRIOR TO REUSE OF THE UNIT.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19 Workmanship. The tractor, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts; cleaning, et cetera.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or

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order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the tractor shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Running-in. The nature and extent of running-in shall be as specified by the manufacturer. In the event running-in is not required, the tractor shall be operated for a distance of 5 miles over improved roads at a speed specified by the contractor.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One tractor shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity. Particular attention shall be given to recording and maintaining a log of the time and man-hours required for all preparation, testing, inspection, and repair. Detailed information shall be provided to indicate compliance (or noncompliance) with all the requirements specified herein.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections

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d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of the following tests:

a. Individual tests

b. Sampling plan and tests.

4.5.1 Individual tests. Each tractor shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Features and components

c. Dimensions and weight

d. Mechanical inspection

e. Functional check.

4.5.2 Sampling plan and tests

4.5.2.1 Periodic sampling. One tractor shall be selected at random from each production run of 50 or less or one from each additional 50 produced and subjected to the following tests as described in 4.6.8.2.1. Where production rates result in less than 50 tractors being produced in a 90-day period, one tractor shall be subjected to the following tests each 90 days:

a. One hundred miles of road operation at 30,000 pounds drawbar pull

b. Fifty miles of figure 8 or continuous S differential testing.

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4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or produced later shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a test failure. But final acceptance of the items on hand or items produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The tractor shall be inspected to determine compliance with the requirements specified herein with respect to material, dimensions, workmanship, and marking.

4.6.2 Features and components. The tractor shall be critically inspected to determine that all necessary components and features are included and properly installed.

4.6.3 Dimensions and weight. The weights and measurements shall be checked; this shall also be performed on major components wherein no disassembly is required, or as directed by the procuring activity. Measurements for overall height ground clearance, towing crew transport, pintle height and clearances, steering wheel clearance and diameter, tool box and chock compartment, tire chain clearance, angle of approach, angle of departure, and overall width of the tractor shall be made.

4.6.4 Mechanical inspection. A critical mechanical inspection shall be made of all components and parts of the tractor and data recorded as to condition, defects of manufacture, and damage through use prior to test. The inspection shall include complete greasing and servicing of the tractor to insure readiness for operation, if required.

4.6.5 Functional check. All components, fixtures, controls, and mechanisms shall be moved through their maximum normal displacements and shall show no malfunctioning or objectionable rough operation, vibration, or irregular movements.

4.6.6 Performance characteristics tests. The tractor shall be tested to determine compliance with the following:

a. Turn radius of 24 feet right and left

b. Maximum cramping angle when turning right or left in oblique and two-wheel steer

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c. Stopping an airplane, simulated airplane or equivalent load, of 750,000 pounds within 35 feet from 5 mph

d. Maximum vehicle speed of 25 mph when not towing an airplane

e. Steering force (pounds force required to turn steering wheel)

f. Pintle chain free falls (see 3.11.4)

g. Drawbar pull of 50,000 pounds.

4.6.7 Gross weight. The tractor shall be weighed to determine compliance with 3.15.

4.6.8 Mobility tests

4.6.8.1 Servicing and adjustments. Servicing or adjustments shall not be permitted during the road tests, except those normally permitted for 500-, 1,000-, and 2,000-mile inspections.

4.6.8.2 Vehicle weight. The tractor shall be equipped to cause a gross weight of not less than 63,000 pounds and shall be subjected to the following tests:

4.6.8.2.1 Road endurance and towing. The tractor shall be subjected to the following tests in the order specified and shall demonstrate performance in a satisfactory manner:

a. 4,000 miles under load as follows. During the tests, the tractor shall be stopped by the service brakes at least once in each 5 miles:

(1) 200 miles at 50,000 pounds drawbar pull

(2) 1,000 miles at 40,000 pounds drawbar pull

(3) 1,600 miles at 30,000 pounds drawbar pull

(4) 800 miles at 20,000 pounds drawbar pull

(5) 400 miles at 12,000 pounds drawbar pull

* b. Proportional wheel drive shall be tested as follows: The tractor shall be coupled to a heavy vehicle or a truck dynamometer capable of applying a steady load which will result in a drawbar pull of not less than 25,000 pounds and not more than 30,000 pounds. The tractor shall be driven on improved roads for a distance of not less than 50 miles while so loaded. During the entire 50 miles, the tractor shall be steered to result in a figure 8 or a continuous S pattern. At the end of this test run, the tractor shall pass the drawbar pull test of 50,000 pounds at least 3 times. Greased

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plates, or equivalent, shall be used under three wheels (tires) with one on concrete or macadam to determine weight to drawbar pull ratio for each tire. Plates shall be rotated until all tires are subjected to this test

c. 9,000 miles of highway operation at no load.

4.6.8.3 Inspection after test. After the endurance tests, a thorough inspection of the tractor shall be accomplished as follows:

a. Major assemblies, such as engine, transmissions, and differential axles, shall be disassembled as necessary to determine the extent of wear and damage. Erratic operation, overheating, excessive noise, or vibration suggesting potential failures shall be indicative that disassembly is necessary

b. All gears, shafts, bearings, parts, and components disassembled shall be subjected to magnetic particle inspection to determine if cracks have developed through stress

c. Castings used to house major assemblies, or used as part of major assemblies, shall be examined under a black light to determine if damage was incurred during the test.

Cracks or excessive wear damage indicating potential malfunction shall constitute failure of the test.

4.6.9 Environmental tests. The tractor shall be subjected to the following environmental tests in accordance with MIL-STD-810:

4.6.9.1 Fungus. Samples of materials used in the construction of the tractor, except for materials specifically designated herein, shall be subjected to fungus-resistance in accordance with method 508, procedure II. In lieu of this test, a critical inspection evaluation may be used to determine component adequacy.

4.6.9.2 High temperature operation. The tractor shall be subjected to high temperature in accordance with method 501, procedure I, except the exposure period shall be 24 hours at a maximum temperature of 125°F. At the end of the 24-hour period and while at this temperature, the tractor shall be started and operated to determine compliance with the performance requirements specified herein. The engine shall be operated at fast idle rpm for a period of 2 hours. There shall be no overheating as a result of this test. During the entire test, periodic checks shall be made to determine that the doors, rubber door gaskets, power cables, hoses, and grease seals are functioning properly.

4.6.9.3 Low temperature operation. The tractor shall be subjected to low temperature in accordance with method 502, procedure I, except the exposure period shall be 24 hours at a temperature of -20°F. At the end of the 24-hour stabilization period, the engine shall be started and operated at rated engine speed and operating horsepower until the coolant temperature of the engine has stabilized at a constant temperature to insure compliance with requirements specified herein. This test shall be run with cab installed and then removed.

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4.6.9.4 Rain. The tractor shall be subjected to rain in accordance with method 506, procedure I. Any leaking of water into the operating components of the tractor shall be cause for rejection if deemed objectionable or detrimental to service use. This test shall be repeated with cab removed and vehicle operated to determine suitability.

4.6.10 Radio interference. The tractor shall be tested for radio interference characteristics in accordance with MIL-STD-826, method 7001.

4.6.11 Winch test. When a winch is installed, an 80,000-pound tensile load, +10 -0 percent shall be applied to the winch cable by the winch on bare drum. The winch shear pins shall fail within the above limits. This test shall be repeated 4 times at the prevailing ambient temperature and once at -20°F. No damage to the cable or winch components shall result, except to the shear pin.

4.6.12 Tiedown attachments. The tractor shall be anchored to a flatcar using only the tiedown attachments. Blocking shall not be used. The flatcar shall be humped with the tractor tied down against a static object (similar to a stop or bumper block) at 11 mph with the tractor facing in the direction of the static object. Five freight cars with their brakes set by air may be used as the static object. The test shall be repeated with the rear of the tractor in the direction of the static object. Weld failure or permanent deformation of any attachment or structural member shall constitute failure of this test.

4.6.13 Reliability demonstration and test. Satisfactory completion of all tests specified herein shall be a demonstration of compliance with the quantitative reliability requirements of this specification.

4.6.14 Maintainability. Maintainability tests shall be conducted to determine the adequacy and ease of preventive maintenance or servicing and ease of performing unit replacement. The removal of one major component in order to service or replace another shall be cause for rejection. Tests shall include, but not be limited to, the following:

a. General tune-up (engine ignition, carburetor, et cetera)

b. Servicing and greasing.

* 4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with MIL-STD-281, level A, B, or C as specified (see 6.2).

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*5.2 Unless otherwise specified, the tractor shall be shipped as a mobile unit.

*5.3 Marking. The tractor shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Tractor, Wheeled, Aircraft Towing A/S32U-30, for 500,000 to 750,000 Pounds Gross Weight Aircraft.

6. NOTES

6.1 Intended use. The A/S32U-30 tractor is intended for use as a conventional means of moving aircraft weighing up to 750,000 pounds gross weight under weather and ramp conditions specified herein.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When the winch assembly is required (see 3.11.5)

*c. When cab is required (see 3.10.1)

d. Location and conditions for preproduction testing (see 4.4).

*e. Level of preservation, packaging, and packing (see 5.1).

6.3 Definitions. For the purpose of this specification, the following definitions apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x}$$

$$\text{where } x = \frac{\text{Mission time}}{\text{MTBF}}$$

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The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees of freedom) distribution even though the failure rate for the test time is zero.

6.3.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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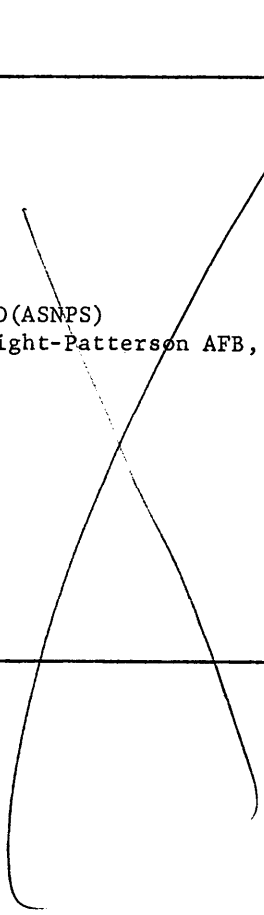
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